

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5:

Measure network performance using key metrics with simulation tools
(BL4,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Total Marks:30

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability
2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.
3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand

ANSWER SHEET

SIMATS ENGINEERING — ASSESSMENT TOOL 1

Course: Computer Networks (CSA07) | CO5: Measure network performance using key metrics with simulation tools

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Question 1: Rural Branch Office Connectivity

Scenario: A multinational company is opening a branch in a remote rural location with no high-speed wired internet. The office must run cloud applications, video conferencing, and secure communication with HQ, on a limited budget, with uninterrupted operations.

1. Understanding of the Problem & Constraints

The core problem is delivering reliable, secure, business-grade connectivity where wired broadband (fibre/DSL) is not available. Four constraints must be balanced simultaneously:

- **Bandwidth:** must sustain simultaneous cloud app usage and HD video conferencing (roughly 4-10 Mbps per active video call).
- **Reliability:** business operations cannot tolerate frequent downtime, so a single point of failure is unacceptable.
- **Cost:** limited budget rules out laying new fibre or leasing a dedicated MPLS line to a remote site.
- **Scalability:** the solution must accommodate future headcount growth without a full redesign.

2. Problem Decomposition

- **Sub-problem A — Last-mile connectivity:** how does the office reach the internet at all?
- **Sub-problem B — Redundancy/failover:** what happens when the primary link fails?
- **Sub-problem C — Local networking:** how are devices inside the office connected?
- **Sub-problem D — Security:** how is traffic to HQ protected over a public/shared medium?
- **Sub-problem E — Future growth:** how is the design extended without replacement?

3. Proposed Solution (Constraints Applied)

A. Internet connectivity

- **Primary link — Satellite broadband (e.g., LEO service such as Starlink Business) or 4G/5G fixed-wireless** since fibre/DSL is physically unavailable. LEO satellite gives 50-200 Mbps with low latency (~25-50 ms), sufficient for video conferencing and SaaS/cloud tools.
- **Secondary/backup link — 4G/5G cellular router** from a different carrier, used purely for automatic failover. This gives path diversity: if the satellite dish is obstructed (weather) or the cellular tower fails, the other path keeps the office online.
- Both links terminate on an **SD-WAN capable router** that performs automatic failover and can also load-balance non-critical traffic across both links, directly satisfying the 'uninterrupted operations' requirement without the cost of a leased line.

B. Networking devices

- **SD-WAN/dual-WAN router** — dual-link failover and traffic shaping (prioritises video/voice over bulk downloads via QoS).
- **Managed Layer-2 switch** — connects office PCs, printers, and VoIP/video units; supports VLAN segmentation for future scaling.

- **Enterprise Wi-Fi Access Point (802.11ac/ax)** — cost-effective wireless coverage for a small branch instead of cabling every desk.
- **UPS (Uninterruptible Power Supply)** — protects against local power instability common in rural areas, supporting reliability.

C. Security mechanisms

- **Site-to-site VPN (IPsec) or SD-WAN native encryption** between branch and HQ — secures all traffic over the satellite/cellular link, which is inherently a shared/public medium.
- **Next-generation firewall (NGFW)** at the branch edge with intrusion prevention (IPS) for threat filtering close to the source.
- **WPA3 encryption + 802.1X authentication** on the Wi-Fi network so only authorised staff devices connect.
- **Multi-factor authentication (MFA)** for access to cloud applications, reducing risk from credential theft.

4. Justification against Constraints

Bandwidth: LEO satellite/5G fixed-wireless provides sufficient throughput for cloud apps and conferencing without the multi-month lead time and expense of trenching fibre. **Reliability:** the dual-WAN + SD-WAN failover design removes the single point of failure, and the UPS covers local power outages. **Cost:** both satellite/5G and SD-WAN routers avoid the capital cost of dedicated leased lines while still meeting business-grade needs — CAPEX is limited to hardware, and OPEX is a monthly service fee. **Scalability:** the managed switch with VLANs and the AP's capacity mean additional staff or a second AP can be added later without re-architecting the network; the SD-WAN router can also add a third WAN link if headcount grows significantly.

5. Creativity/Innovation Note

Using LEO satellite broadband as a primary (not just backup) link is a modern alternative to traditional VSAT, offering far lower latency — making it realistic for real-time video conferencing, which older satellite technology could not support.

Question 2: Campus-Wide Wireless Network Design

Scenario: A university needs campus-wide Wi-Fi for 3,000+ students across classrooms, labs, libraries, and hostels, with a limited number of access points (APs), minimal interference, and secure access.

1. Understanding of the Problem & Constraints

- **Scale:** 3,000+ simultaneous/near-simultaneous users across diverse zones.
- **Limited AP count:** budget restricts the number of physical access points, so each AP must be used efficiently.
- **Interference:** overlapping signals between adjacent APs (co-channel interference) degrade performance if not planned.
- **Security:** only authorised students/staff should access the network.
- **Cost vs coverage vs performance trade-off** across very different zone types (dense classrooms vs. sparse hostel corridors).

2. Problem Decomposition

- **Sub-problem A — Zone classification:** group areas by user density and usage pattern (high-density: classrooms/labs/library; medium: hostels; low: open outdoor areas).
- **Sub-problem B — AP placement & capacity planning:** how many APs per zone and where.
- **Sub-problem C — Frequency/channel planning:** which band and channels minimise interference.
- **Sub-problem D — Authentication & access control:** how users are verified.
- **Sub-problem E — Cost control:** matching AP count to budget while meeting coverage.

3. Proposed Wireless Network Design

A. Access point placement

- **High-density areas (classrooms, labs, library):** ceiling-mounted APs at a density of roughly one AP per 25-30 users / ~150-200 sq. m, using directional/patch antennas to focus coverage and reduce spill-over into neighbouring rooms.
- **Hostels (medium density, many walls):** one AP per floor/wing corridor rather than per room, using omnidirectional antennas; drywall/concrete attenuation is exploited to naturally reduce cross-floor interference.
- **Outdoor/common areas (quad, canteen):** fewer, higher-power outdoor-rated APs mounted on poles for wide coverage since user density is lower and roaming tolerance is higher.
- AP placement follows a **cellular/honeycomb layout** with slight coverage overlap (~10-15%) for seamless roaming, avoiding large overlaps that waste AP budget.

B. Frequency band selection

- **Dual-band operation (2.4 GHz + 5 GHz), migrating to Wi-Fi 6/6E (2.4/5/6 GHz)** where budget allows.
- **5 GHz as primary band in high-density zones** — more non-overlapping channels (36-165) allow adjacent APs to be assigned different channels, minimising co-channel interference; shorter range also naturally limits spill-over into other classrooms.
- **2.4 GHz reserved for range-limited devices/legacy IoT** and low-density outdoor zones, using only the 3 non-overlapping channels (1, 6, 11) with careful channel reuse planning.
- **Wi-Fi 6 (802.11ax)** is preferred where budget permits because OFDMA lets a single AP serve many simultaneous low-bandwidth clients (typical of classrooms) far more efficiently than older standards — directly addressing the 'limited AP count' constraint.

C. Authentication methods

- **WPA2/WPA3-Enterprise with 802.1X** tied to the university's RADIUS server, so each student/staff logs in with their institutional credentials (no shared passwords, individual accountability).
- **eduroam-style federated authentication** for visiting researchers/students from partner institutions.
- **Captive portal + guest VLAN** for short-term visitors, isolated from the academic network.
- **Network Access Control (NAC)** to restrict device types (e.g., limit IoT devices to a separate VLAN).

4. Justification — Balancing Coverage, Performance, Security, Cost

Zone-based AP density (rather than uniform spacing) concentrates the limited AP budget where user density is highest, maximising coverage-per-rupee. 5 GHz-first channel planning with a honeycomb layout minimises interference so each AP delivers consistent throughput despite fewer total units. WPA2/3-Enterprise with RADIUS gives strong, centrally-managed security at near-zero marginal cost per user (leverages existing student credentials) rather than expensive per-device certificates. Overall, the design meets the coverage need for 3,000+ users while keeping AP count — and therefore capital cost — as low as the density and channel planning will allow.

Question 3: Secure Network Architecture for a Financial Institution

Scenario: A financial institution must upgrade internal network security to protect customer data and online banking while complying with regulations, without significantly increasing cost or hurting performance.

1. Understanding of the Problem & Constraints

- **Security:** must protect highly sensitive financial/customer data from internal and external threats.
- **Performance:** security controls (firewalls, IDS, encryption) must not introduce unacceptable latency to banking transactions.
- **Compliance:** must meet regulatory standards (e.g., PCI-DSS for card data, RBI/regional banking cybersecurity frameworks) which mandate segmentation, logging, and access control.
- **Budget:** the redesign must reuse/upgrade existing infrastructure where possible rather than a costly rip-and-replace.

2. Problem Decomposition

- **Sub-problem A — Network segmentation:** how to isolate sensitive systems from general traffic.
- **Sub-problem B — Perimeter & internal firewalling:** where firewalls are placed and what they inspect.
- **Sub-problem C — Authentication:** how staff and customers are verified before reaching sensitive systems.
- **Sub-problem D — Threat detection:** how intrusions/anomalies are identified in real time.
- **Sub-problem E — Compliance mapping:** ensuring each control satisfies an auditable regulatory requirement.

3. Proposed Secure Architecture

A. Network segmentation

- **VLAN-based segmentation** separating: (1) core banking/database servers, (2) employee workstations, (3) customer-facing web/online-banking servers in a **DMZ**, (4) guest/administrative Wi-Fi.
- This limits lateral movement — a compromised employee laptop cannot directly reach the core banking database — which is both a security and a PCI-DSS/compliance requirement.
- **Micro-segmentation** for the database tier storing customer records, with access restricted to specific application servers only.

B. Firewall deployment

- **Perimeter NGFW** between the internet and the DMZ, doing stateful inspection, SSL/TLS decryption for threat scanning, and application-layer filtering.
- **Internal firewall** between the DMZ/employee VLAN and the core banking segment — a second layer so that even if the DMZ is breached, the core segment is separately defended (defence-in-depth).
- Firewall rules follow the **principle of least privilege** (deny-all default, explicit allow rules only) to keep the rule set small and auditable, controlling operational cost.

C. Authentication methods

- **Multi-factor authentication (MFA)** for all employee access to core banking systems and for customer online banking login.
- **Role-Based Access Control (RBAC)** so staff only see data required for their role (teller vs. loan officer vs. admin) — directly supports compliance audit trails.

- **Certificate-based authentication** for server-to-server communication between application and database tiers.

D. Intrusion detection/prevention

- **Network IDS/IPS** at the DMZ and core-segment boundary to flag/block known attack signatures and anomalous traffic in real time.
- **SIEM (Security Information and Event Management)** aggregating logs from firewalls, IDS, and servers for correlation, alerting, and the audit trail regulators require.
- **Regular vulnerability scanning** as a low-cost, ongoing compliance control rather than expensive continuous penetration testing.

4. Justification — Security, Performance, Compliance, Budget

Segmentation contains breaches without needing new hardware beyond VLAN-capable switches already common in enterprise networks, keeping cost low. Placing the NGFW only at trust boundaries (perimeter and core-segment edge) — rather than inspecting all traffic everywhere — limits the performance overhead to the points where it matters most. MFA and RBAC are largely software/licensing costs, not new hardware, keeping budget impact low while directly satisfying compliance mandates for access control and audit logging. The SIEM consolidates compliance evidence (logs, alerts) into one system, reducing the manual effort — and therefore cost — of audit preparation.

5. Summary Table — Constraint Mapping

Constraint	How it is Addressed
Security	VLAN segmentation, dual-layer NGFW, IDS/IPS, MFA, RBAC
Performance	Firewalls only at trust boundaries; SSL offload; segmentation reduces broadcast/scan traffic
Compliance	Segmentation + logging + RBAC map directly to PCI-DSS/banking security audit requirements
Budget	Reuses VLAN-capable switching; MFA/RBAC are largely software cost; no full network replacement